Anika Tabassum

Mobile: +447466513135 **Email:** tabassum.bidisha@gmail.com **Address:** United Kingdom
Github

EDUCATION

MSc Robotics and Artificial Intelligence **Sep 2025 – Sep 2026**
Queen Mary University of London

- **Key Modules:** Cognitive Robotics, Machine Learning and Artificial Intelligence, Medical Robotics and Surgical Techniques, Deep Learning for Data and Image Analysis
- **Key Projects:** Electro-optical PPG system, Few-Shot Learning in Medical Imaging

BSc Civil Engineering **Jan 2016 – Mar 2021**
Rajshahi University of Engineering & Technology

- **Key Modules:** Mathematics I and II, Physics I and II, Numerical Methods & Computer Programming, Basic Electrical Engineering.
- **Key Projects:** Design of Water Supply and Sewerage Systems, Structural Design of Reinforced Concrete and Balanced Cantilever Bridges, Hydraulic Structure (Weir / Barrage) and Groundwater Resource Design.

TECHNICAL SKILLS

Hardware & Benchtop Testing: Breadboard Prototyping, Commissioning, Continuity Testing, Sensor Calibration, Arduino, PPG Circuits.
Programming & Computing: Python, C++, C, MATLAB, JavaScript.
Machine Learning & Modelling: Physics-Structured ML, Neural Networks (U-Net, MLP), Few-Shot Learning (Reptile), ANFIS, PCA.
Engineering Design & CAD: Systems Requirements, AutoCAD, SketchUp Pro, ETABS, STAAD Pro, Material Selection.

PROJECTS

Physics-Structured sEMG Grip-Force Estimation and Electrode Reduction

- Developed a proof-of-concept physics-structured ML framework for subject-independent sEMG grip-force estimation, maturing theoretical concepts into an application-ready electrode-reduction pipeline.
- Executed a five-fold cross-validation testing programme on Ninapro DB2, achieving a mean held-out R^2 of 0.569 vs. 0.335 for the Random Forest baseline.
- Optimised hardware requirements by reducing 12 sEMG channels to 6 electrodes with only a 0.019 drop in R^2 (within the 0.03 tolerance).

Electro-Optical PPG Sensor Rig: Benchtop Assembly, Testing & Calibration

- Assembled, wired, and commissioned a benchtop breadboard prototype for an electro-optical PPG sensing system from technical schematics in a collaborative team.
- Conducted hardware diagnostics, continuity testing, and benchtop calibration by varying circuit resistance; performed empirical baseline calibration on proximity and ultrasonic sensors via physical reference standards.

Universal Mobility Suite (Airbus Defence and Space Competition Finalist)

- Led conceptual R&D of a space-rated lower-limb exoskeleton for an open-ended lunar mobility challenge, guiding the multidisciplinary proof-of-concept to the competition finals.
- Defined material and actuator requirements (carbon-fibre composite, Ti 6Al-4V, series elastic actuators) and formulated system-level power budgets (2.1 kWh battery), projecting a 50% metabolic demand reduction.

Utilization of Plastic Wastes in Road Construction (Material R&D Literature Review)

- Conducted a comprehensive technical feasibility study to evaluate the technology maturation of polymer waste as a sustainable binder in flexible pavements, addressing open-ended environmental and structural durability challenges.
- Synthesised and analysed experimental lab-scale testing data from literature across multiple polymers, evaluating their thermal characteristics and impact on Marshall Stability and structural flow values.
- Determined that integrating 6% to 8% high-density polyethylene (HDPE) waste optimally enhances road durability; communicated technical findings clearly in a peer-reviewed publication.
- **Publication:** Rashid, G. M. H., Tabassum, A. & Mahfuj, R., "A Review Report on the Utilization of Plastic Wastes in Road Construction", Journal of Transportation Systems, Volume 6, Issue 2, 2021, 1-8.

Post-Disaster Shelter Model Construction

- Defined structural and operational requirements for resilient emergency housing, developing a conceptual proof-of-concept shelter model for rapid deployment.
- Balanced rapid assembly constraints with long-term environmental sustainability to deliver a robust engineering framework; published the resulting methodology in a peer-reviewed environmental engineering journal.
- **Publication:** Rashid, G. M. H., Mahfuj, R. & Tabassum, A., "A Paradoxical Approach to Construct Post Disaster Shelter Model", Journal of Environmental Engineering and Studies, Volume-6, Issue-1, 2021, 39-49.

WORK EXPERIENCE

STEM Instructor – Udvash Coaching Center **Jan 2022 – May 2023**

- Tasked with preparing diverse cohorts for competitive engineering entrance exams, communicated complex mathematics and physics concepts clearly to both technical and non-technical audiences, resulting in improved student comprehension and strong analytical pass rates.
- Collaborated effectively with senior colleagues to plan, organise, and prioritise intensive lecture schedules and mock-testing programmes under tight timelines, successfully delivering structured academic support across multiple batches.
- Exercised people leadership in an inclusive classroom environment by adapting teaching methodologies in real time to troubleshoot open-ended analytical problems, building student confidence in tackling unfamiliar STEM challenges.

CERTIFICATIONS

- **The Complete Python Bootcamp from Zero to Hero in Python** – Udemy
- **Web Design and Development** – Texlab IT
- **Internet of Things (IOT)** – Gobeshona Learning Academy
- **AI in the Workplace** – Northeastern University

EXTRACURRICULARS

- Volunteered for several years with Quantum Foundation Bangladesh, supporting community welfare and humanitarian initiatives.
- Organised donations for low-income families affected by the Covid-19 pandemic.
- Served as a member of the Astronomy & Science Society of RUET club; participated in the 6th Bangladesh Chemistry Olympiad.
- Participated in the 'Developing the Reading Habit Program' arranged by the World-Literature Centre (Bishwo Shahitto Kendro).